

Lemma 1 is used in Eq. 41 which implies $\mathbf{u}^{(l)} \rightarrow \mathbf{u}^*$, $\frac{\mathbf{v}^{(l)}}{\mathbf{q}^{(l)}} \rightarrow \frac{\mathbf{v}^*}{\mathbf{q}^*}$.
Furthermore,

$$\begin{aligned} d_{\mathcal{H}}(\mathbf{v}^{(l+1)}, \mathbf{v}^*) &= d_{\mathcal{H}}\left(\frac{\mathbf{q}^{(l)} + \mathbf{d}}{\mathbf{K}^{(l)}\mathbf{u}^{(l+1)}}, \frac{\mathbf{q}^* - \mathbf{s} + \mathbf{d}}{\mathbf{K}^*\mathbf{u}^*}\right) \\ &= d_{\mathcal{H}}\left(\frac{\mathbf{K}^{(l)}\mathbf{u}^{(l+1)}}{\mathbf{q}^{(l)} + \mathbf{d}}, \frac{\mathbf{K}^*\mathbf{u}^*}{\mathbf{q}^* - \mathbf{s} + \mathbf{d}}\right) \\ &\leq \lambda(\mathbf{K})d_{\mathcal{H}}\left(\frac{\mathbf{u}^{(l+1)}}{\mathbf{q}^{(l)} + \mathbf{d}}, \frac{\mathbf{u}^*}{\mathbf{q}^* - \mathbf{s} + \mathbf{d}}\right) \end{aligned} \quad (42)$$

Lemma 1 is used again in Eq. 42 which shows that $\mathbf{v}^{(l)} \rightarrow \mathbf{v}^*$, $\frac{\mathbf{u}^{(l)}}{\mathbf{q}^{(l)}} \rightarrow \frac{\mathbf{u}^*}{\mathbf{q}^*}$.
Substituting $\mathbf{u}^*$ and $\mathbf{v}^*$ into Eq. 34, which deduces to

$$\text{Diag}(\mathbf{K}^*) = \frac{\mathbf{d}}{\mathbf{u}^* \odot \mathbf{v}^*}, \quad \mathbf{q}^* = \frac{\mathbf{s}}{2} + \left((\mathbf{K}^*\mathbf{v}^*) \odot ((\mathbf{K}^\top)^*\mathbf{u}^*) + \frac{\mathbf{s}^2}{4} \right)^{\frac{1}{2}} - \mathbf{d}. \quad (43)$$

□